SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

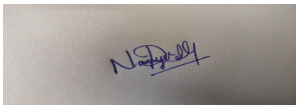
CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Case Study

Weightage: 10%

Code of Conduct:

I NARRAVULA NAVYA SRI - 192324220 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

1. Correct HTML tag for video embedding

Correct option: B. `<video src="video.mp4" controls></video>`

Justification: The HTML5 `<video>` element is used to embed video content in a webpage. The `src` attribute specifies the video file, while `controls` displays playback controls such as play, pause, and volume.

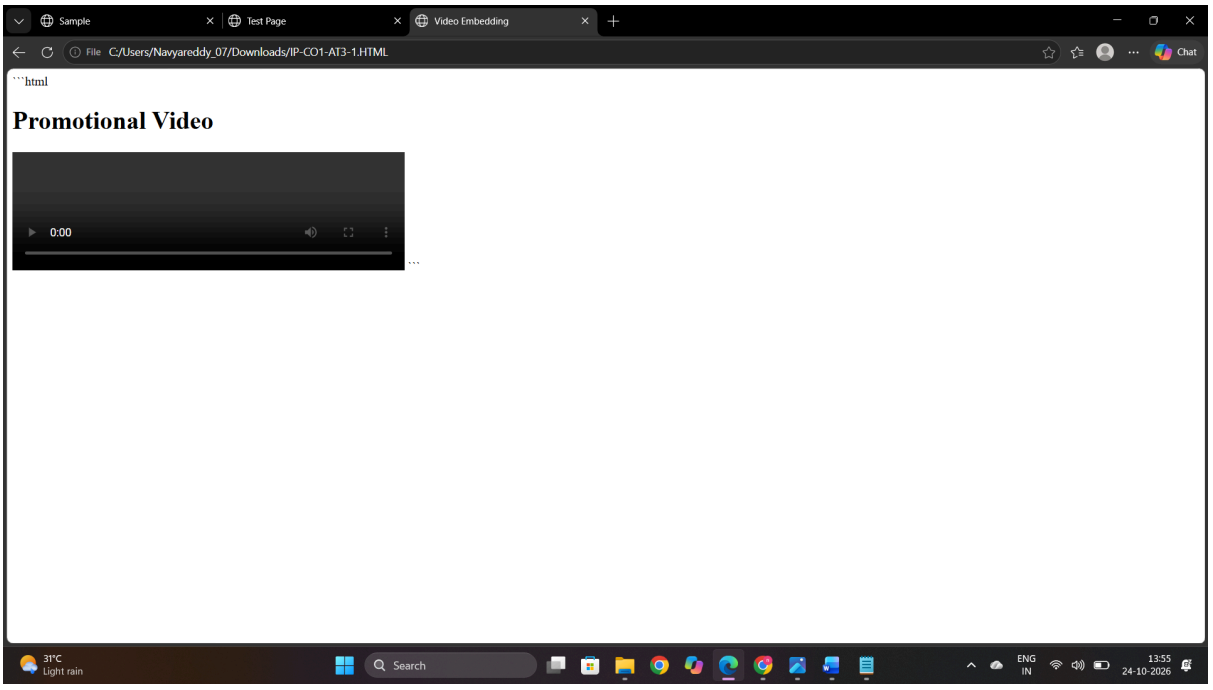
Example:

<video src="video.mp4" controls>

Your browser does not support HTML video.

</video>

The other options (`<media>`, `<movie>`, and `<vid>`) are not standard HTML elements for embedding video.



2. Appropriate HTTP method for form submission

Correct option: B. POST

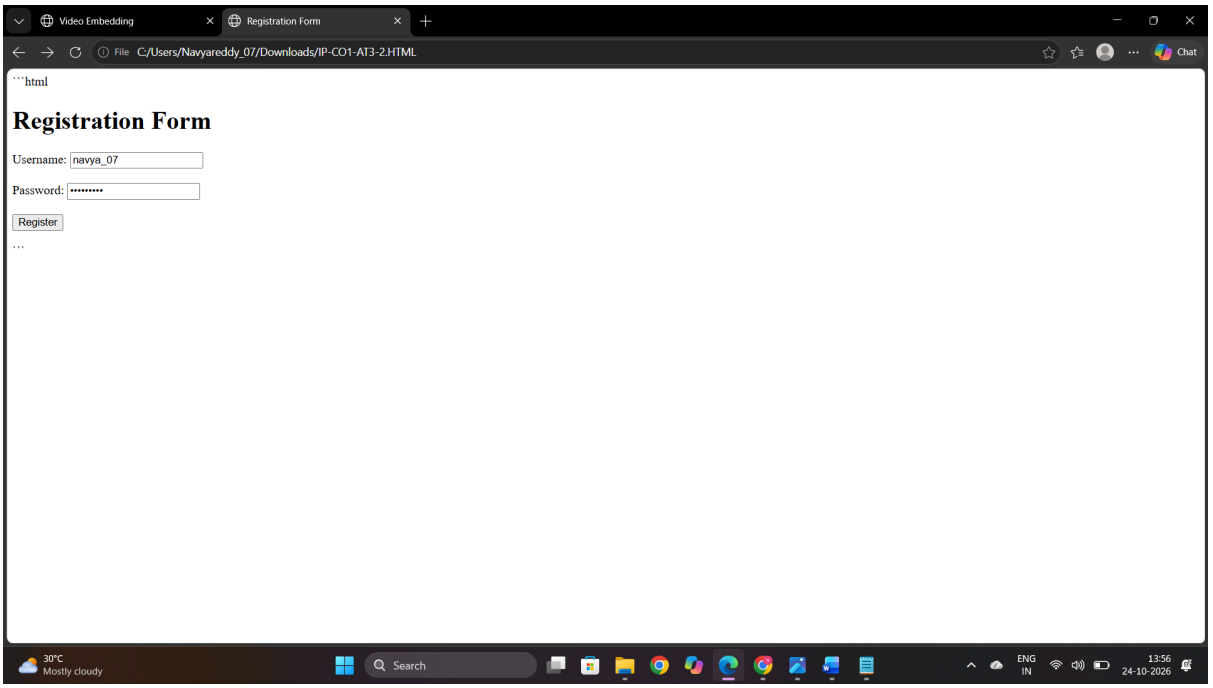
Justification: The POST method sends form data in the request body, making it suitable for submitting registration details to a server. Unlike GET, it does not normally place submitted values in the URL.

Example:

```
<form action="/register" method="POST">
  <label for="username">Username:</label>
  <input type="text" id="username" name="username" required>
  <label for="password">Password:</label>
  <input type="password" id="password" name="password" required>
  <button type="submit">Register</button>
</form>
```

Important: POST alone does not provide encryption. Use HTTPS to protect sensitive data during transmission and validate the submitted data on the server.

- GET: Primarily retrieves resources.
- POST: Submits data for processing.
- PUT: Creates or replaces a resource at a specified URI.
- DELETE: Requests deletion of a resource.



3. Proper semantic element for navigation

Correct option: B. `<nav>`

Justification: The HTML `<nav>` element represents a section of a webpage containing major navigation links. It improves document semantics and helps assistive technologies identify navigation regions.

Example:

<nav>

  Home

  About

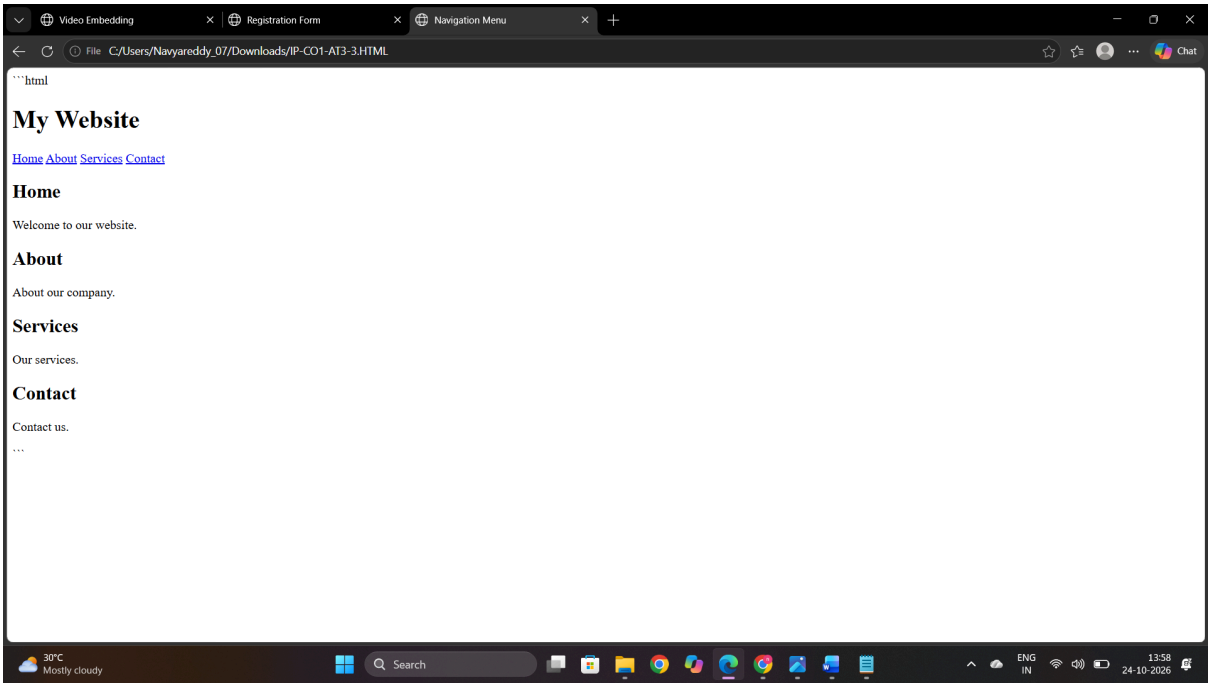
  Services

  Contact

</nav>

The other options have different purposes:

- `<div>`: A generic container without inherent semantic meaning.
- `<section>`: Groups thematically related content.
- `<header>`: Represents introductory content or a section's header.



4. Correct XML syntax

Correct option: B.

<note>

<to>User</to>

<from>Admin</from>

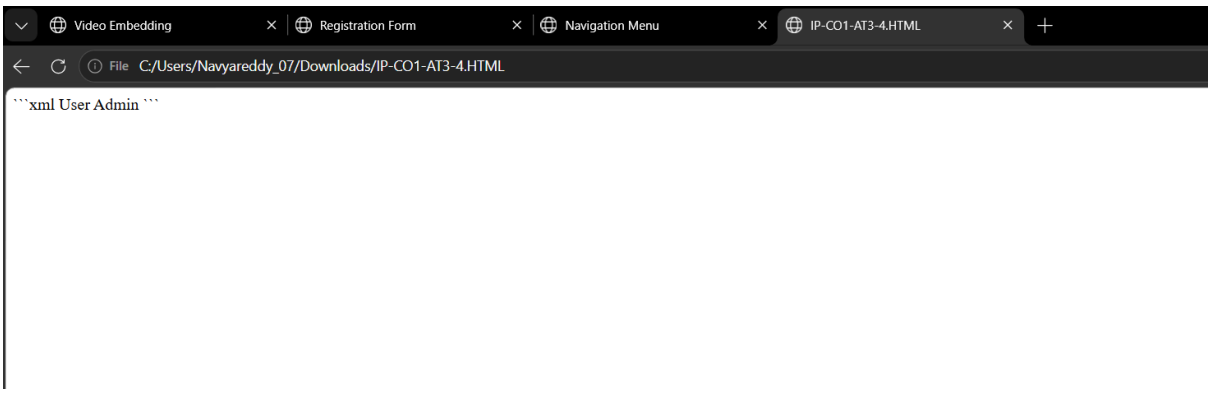
</note>

Justification: XML requires properly nested elements, matching opening and closing tags, and exactly one root element in a well-formed XML document. Option B satisfies these requirements for the given snippet.

Explanation of the other options:

Option	**Error**
A	The root <note> element is not closed.
B	Correctly nested elements with matching closing tags.
C	The <to> element is incorrectly closed with </from>.
D	The <note> closing tag is incomplete; it must be </note>.

XML is case-sensitive, so tag names must match exactly. Attributes must also be quoted correctly.



5. Correct protocol for web communication

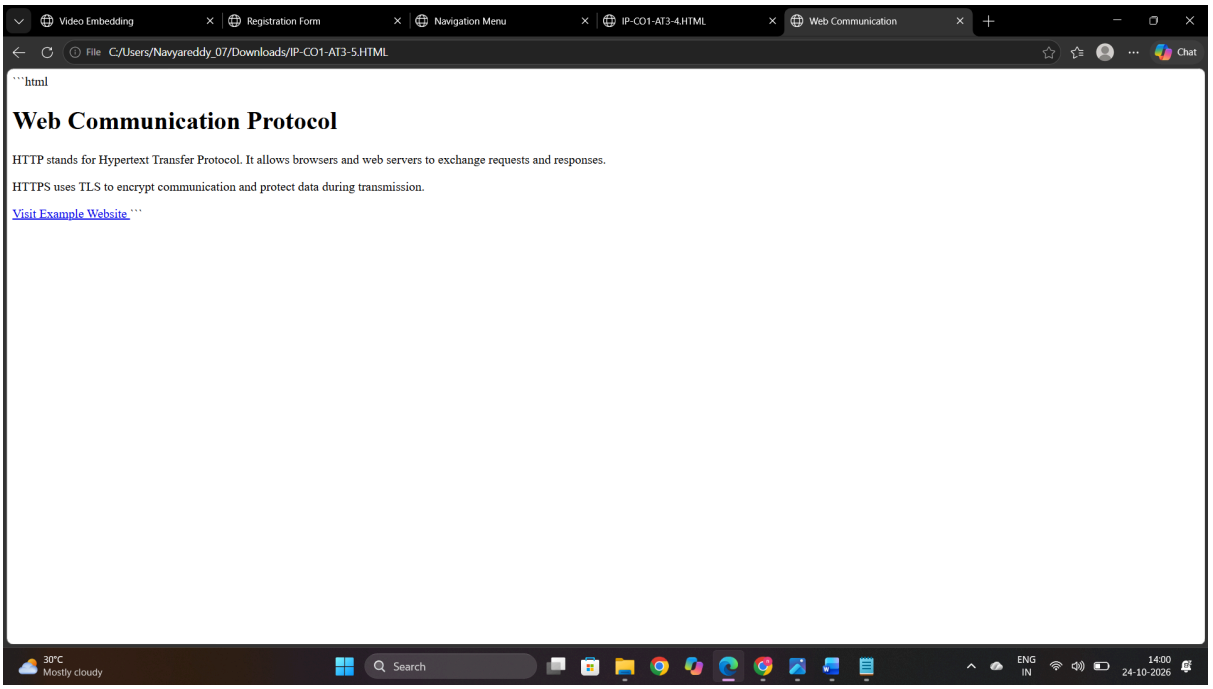
Correct option: B. HTTP

Justification: HTTP (Hypertext Transfer Protocol) is used for communication between web clients, such as browsers, and web servers. It defines how requests and responses are exchanged.

HTTPS is the secure version of HTTP, using TLS to protect communication through encryption and authentication.

The other protocols serve different purposes:

- FTP: Transfers files between systems.
- SMTP: Sends email.
- SNMP: Monitors and manages network devices.



Final answer summary

Question	Correct option	Answer
1. Video embedding	B	<video>
2. Form submission	B	POST
3. Navigation element	B	<nav>
4. XML syntax	B	Properly nested and closed XML elements
5. Web communication	B	HTTP

Conclusion: Following HTML semantic standards, choosing appropriate HTTP methods, maintaining well-formed XML, and using HTTP/HTTPS correctly helps create reliable, accessible, and secure web applications.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand